Introduction to t-testing

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Summary

A t-test is a statistical test used to compare the means of samples. It is a hypothesis test with a test statistic that follows a t-distribution. This guide will focus on two different types of t-test, a Student's t-test and a Welch's t-test. It will explain when each one is used and show you how construct them and use them in practice.

Before reading this guide, it is recommended that you read Guide: Introduction to hypothesis testing, Guide: Introduction to probability, Guide: Expected value, variance, standard deviation, and Factsheet: t-distribution

A *t-test* is a commonly used statistical test for comparing sample means. A t-test is used with a sample to make inferences about a population. It helps determine whether the sample provides enough evidence to draw conclusions about the population mean. In this guide, μ represents the population mean that you are estimating and testing based on the sample.

This guide will cover two types of t-tests: **Student's t-test** and **Welch's t-test**. The origin of Student's t-test is really interesting! In 1908, William Sealy Gosset was working as a statistician at the Guinness Brewery in Dublin, Ireland. His job was to monitor quality control in the brewing process. Unlike many statisticians at the time, Gosset often had to make decisions using very small sample sizes. While looking at these small samples, Gosset noticed something important: the means and variances calculated from small samples did not behave according to the normal distribution, which most statisticians relied on. Realising that the standard approach was not suitable for small samples, he developed a new probability distribution that accounted for this extra uncertainty. This became known as a Student's t-distribution, and it eventually led to the Student's t-test. But why is it called the "Student's" t-test? Because the Guinness Brewery did not want its sampling methods leaked to the public. So, Gosset published his work under the pen name "Student". Later, another statistician, Bernard Lewis Welch, extended Gosset's work. He developed what is now called Welch's t-test, which considers the original method and also makes comparisons between two samples when their variances are unequal.

What is a t-test?

Definition of a t-test

A t-**test** is a hypothesis test where the test statistic follows a t-distribution.

For more on hypothesis testing, see Guide: Introduction to hypothesis testing and for more on t-distributions, see Factsheet: t-distribution.

Tip

In statistics, a t-test is typically used when the sample size is less than 30. With small samples, the estimate of the standard deviation is less stable. As the sample size increases, the estimate of the standard deviation becomes more stable, and the t-distribution gradually approaches the normal distribution. As the sample size reaches 30, the difference between the t-distribution and the normal distribution becomes insignificant. So, a common *rule of thumb* in statistics is to use the t-distribution when the sample size is *less than* 30.

Example 1

Cantor's Confectionery wants to test the average weight of their best selling chocolate swirls. They gather a sample of 21 swirls and weigh each one, calculating the sample mean weight.
You can use a t-test to do this.

What is a Student's t-test?

Definition of Student's t-test

A **Student's** t-**test** is a t-test where the samples are known to, or assumed to, have equal variance (σ^2).

When do you use a Student's t-test?

When you have one sample

A **one-sample Student's** t-**test** is a t-test where you have a single sample with one sample mean. The test involves comparing the sample mean to a known value.

Example 2

Cantor's Confectionery sell a 24 door advent calendar, behind each door is a chocolate swirl. They open up one advent calendar and weigh all 24 swirls. The sample mean is 12.1 grams. They want to test if the average weight of a chocolate swirl is equal to 13 grams, as stated on their website.

You have a small sample of 24 chocolate swirls, which has a sample mean of 12.1 grams. You are testing the sample mean against a known value of 13 grams. So, you can use a Student's t-test to do this.

When you have two samples with equal variance

A **two-sample Student's t-test** is a t-test where you have two samples, so two sample means. The test involves comparing the two sample means to **one another**.

Example 3

Cantor's Confectionery sells a vegan alternative to their 24 door advent calendar. They open one vegan calendar and calculate the sample mean weight of the swirls, then do the same for a regular chocolate calendar. They want to test if the average weight of the vegan swirls is the same as the average weight of the regular swirls, assuming both samples have equal variance.

You have two samples, each of size 24 and with equal variance. So, you can use a two-sample Student's t-test to do this.

When you have paired samples

A **paired Student's t-test** is a t-test that is used when you measure the same sample twice. The test involves comparing two sample means, taken from the same sample. It is most commonly used when sampling before and after a change to the sample. For example, before and after packaging.

Example 4

During the festive season, Cantor's Confectionery bring out their snowball caramel apples (which are their regular caramel apples... dipped in white chocolate). They take a sample of 15 caramel apples and calculate the sample mean weight. They then dip the caramel apples in chocolate and re-calculate the sample mean weight. The samples are assumed to have equal variance.
You have two samples which are paired and have equal variance. So, you can use a paired Student's t-test to test the weight of the apples before and after being dipped in chocolate.

Summary: When to use a Student's t-test

- When you are comparing one sample mean to a known value.
- When you are comparing two sample means, with equal variance, to each other.
- When you are comparing two paired sample means, with equal variance, to each other.

What is a Welch's t-test?

Definition of Welch's t-test

A Welch's t-test is a t-test where the samples have unequal or unknown variances (σ_1^2) and (σ_2^2).

When do you use a Welch's t-test?

When you have two samples with unequal variance

A **Welch's t-test** must always be a two sample test. This is because Welch's t-test is used on two samples with unequal variances.

Example 5

Cantor's Confectionery are partnering with Dirichlet's Distributors to mass produce chocolate swirls. They open two new factories and want to make sure that the average weight of a chocolate swirl is the same in both factories. They collect two batches, both of sample size 25. The sample means and sample variances are different from both factories.

You have two samples with unequal variance, so you can use a Welch's t-test to do this.

Summary: When to use a Welch's t-test

- When you are comparing two sample means, with unequal or unknown variances, to each other

What other assumptions are made when constructing a t-test?

Independence

All types of t-test require the observations to be **independent**. So, within the sample itself, the observations must be **independent**. Also, for two samples, the observations between samples must be **independent** too.

Random Sampling

t-testing assumes that all of the data used is collected using a random sampling method. Typically, this method is **simple random sampling**, meaning that all of the data is selected randomly.

For more information on **simple random sampling**, see Factsheet: Simple random sampling.

One and two-tailed t-tests

Before constructing a t-test you also must decide whether you are performing a **one or two-tailed test**.

- If you want to know whether two sample means are equal or different from each other, perform a **two-tailed test.**

Example 6

Cantor's Confectionery wants to test if their profits are the same on weekdays and weekends. To do this, they take a sample of profits from 2 random weekdays and the 2 weekend days, then calculate the sample means.
Since you are testing whether the sample means are equal, this is a **two-tailed test.**

- If you want to know whether one sample mean is greater or less than the other sample mean, perform a **one-tailed test**.

Example 7

Cantor's Confectionery wants to test if their profits are higher on weekends than on weekdays. To do this, they take a sample of profits from 2 random weekdays and the 2 weekend days, then calculate the sample means.
Since you are testing whether the weekend profits are greater than the weekday profits, this is a **one-tailed test**.

For more about one and two tailed tests and why they are used, see: Guide: One and two-tailed tests.

How to construct a t-test

Step 1: What type of t-test?

You can consider whether you are looking to perform:

- A Student's or Welch's t-test?
- A one or two-tailed t-test?

t-test flowchart

Which t-test should I use?

Step 1: *How many samples do you have?*

- One sample: go to Step 2
- Two samples: go to Step 3

Step 2: One-tailed or two-tailed?

- One-tailed: You have a one-sample Student's t-test (one-tailed)
- Two-tailed: You have a one-sample Student's t-test (two-tailed)

Step 3: Paired or independent? Paired: Paired-samples Student's t-test (can be one-tailed or two-tailed) Independent: go to Step 4

Step 4:Equal variances? Yes: Independent-samples Student's t-test No: Welch's t-test

Step 2: Define your null and alternative hypothesis

H_0 is your **null hypothesis.** This is what you believe to be to be true. H_1 is your **alternative hypothesis**. This is the alternative case which you are testing for.

> **Example 8**
>
> Cantor's Confectionery takes a sample of 10 sugar cookies and calculates their average diameter to determine how much sugar to purchase for next month. This month, they assumed the average diameter was 9cm but now they want to test if the average diameter is greater than 9cm.
> You can construct a null and alternative hypothesis for this.
>
> - This is a one-sample Student's t-test.
> - This is a one-tailed t-test.
>
> So, your hypothesis is $H_0 : \mu = 9 \quad H_1 : \mu > 9$.
> Cantor's Confectionery take a second sample of 10 sugar cookies and calculate their average diameter. They want to test whether this sample mean is the same as the first sample mean. They assume equal variance between samples.
> You can construct a null and alternative hypothesis for this.
>
> - This is a two-sample Student's t-test.
>
> - This is a two-tailed t-test.
>
> So, your hypothesis is $H_0 : \mu_1 = \mu_2 \quad H_1 : \mu_1 \neq \mu_2$.

Example 9

A rival company, Lovelace's Lollies, believes their sugar cookies have a wider diameter than those made at Cantor's Confectionery. They take a sample of 10 sugar cookies and calculate the average diameter. They assume equal variance between their sample and Cantor's Confectionery's sample.
You can construct a null and alternative hypothesis for this.

- This is a two-sample Student's t-test.
- This is a one-tailed t-test.

So, your hypothesis is $H_0 : \mu_1 = \mu_2 \quad H_1 : \mu_1 > \mu_2$.

Step 3: Define your significance level (α)

For all types of hypothesis testing, including t-testing. you need to choose a significance level (α). In statistical testing, it is common to see $\alpha = 0.01$, $\alpha = 0.05$, and $\alpha = 0.1$.

Fore more on significance levels, see Guide: Introduction to hypothesis testing.

Step 4: Calculate your test statistic (τ)

The **test statistic** is used to tell you how far away your result is from the null hypothesis. How you calculate τ depends on the type of test you are conducting and the number of samples you have.

What does the test statistic tell you?

- If your test statistic is **small**, it suggests that your result is **close** to what was predicted in your null hypothesis.
- If your test statistic is **large**, it suggests that your result is **far** to what was predicted in your null hypothesis.

"Small" and "large" are always judged relative to the critical value of your t-test.

One-sample test statistic

For a **one-sample t-test,** where:

- $\bar{x}$ is the sample mean.

- μ_0 is the particular value you are comparing the sample mean to.
- s is the sample standard deviation.
- n is the sample size.

then, the t-**test statistic** (τ) is:

$$\tau = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

! Important

Often, $s/\sqrt{n}$ is referred to as the **standard error** of the sample.

Example 10

Cantor's Confectionery want to compare the average weight of their chocolate Easter eggs to the market average. They took a sample of 13 Easter eggs and calculated the sample mean weight. They now need to calculate a test statistic using the data they have collected.

Quantity	Symbol	Value and unit
Sample mean	$\bar{x}$	92 grams
Market average	μ_0	88.5 grams
Sample standard deviation	s	2.3 grams
Sample number	n	13

The test statistic is given by

$$\tau = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

so

$$\begin{aligned}\tau &= \frac{92 - 88.5}{2.3/\sqrt{13}} \\ &= 5.4867 \qquad \text{to 4dp.}\end{aligned}$$

Two-sample test statistic

For a **two-sample t-test,** where:

- μ_1 is the sample mean of the first sample.
- μ_2 is the sample mean of the second sample.
- s_1 is the sample standard deviation of the first sample.
- s_2 is the sample standard deviation of the second sample.
- n_1 is the sample size of the first sample.
- n_2 is the sample size of the second sample.

then, the **t-test statistic** (τ) is:

$$\tau = \frac{\mu_1 - \mu_2}{\sqrt{\left(\frac{s_1^2}{n_1}\right) + \left(\frac{s_2^2}{n_2}\right)}}$$

! Important

This test statistic will look different for a Student's t-test and a Welch's t-test. For a Student's t-test, s_1 and s_2 are the same. For a Welch's t-test, they are not.

Example 11

Lovelace's Lollies have taken two samples of their bestselling lollipops. For both samples, they weighed each lollipop and calculated the sample mean weight. They want to calculate a test statistic so that they can compare the two sample means. Their computer system produced the output below.

Quantity	Symbol	Value and unit
Sample 1 mean weight	μ_1	11 grams
Sample 2 mean weight	μ_2	12 grams
Sample 1 standard deviation	s_1	0.49 grams
Sample 1 size	n_1	15
Sample 2 standard deviation	s_2	0.49 grams
Sample 2 size	n_2	18

The standard deviation is the square root of the variance. Sample one and two have the same standard deviation so equal variance. The test statistic is given by:

$$\tau = \frac{\mu_1 - \mu_2}{\sqrt{(\frac{s_1^2}{n_1}) + (\frac{s_2^2}{n_2})}}$$

so

$$\tau = \frac{11 - 12}{\sqrt{(\frac{0.49^2}{15}) + (\frac{0.49^2}{18})}} = -5.8375 \qquad \text{to 4dp.}$$

Important

A test statistic can be negative! The sign of a test statistic depends on the nature of the test and the data you have.

Example 12

Cantor's Confectionery now take a second sample of Easter eggs on another day to compare to their first sample in Example 10. They want to produce a test statistic with the following computer output:

Quantity	Symbol	Value and unit
Sample 1 mean	μ_1	92 grams
Sample 2 mean	μ_2	91.45 grams
Sample 1 standard deviation	s_1	0.4 grams
Sample 2 standard deviation	s_2	0.25 grams
Sample 1 size	n_1	23
Sample 2 size	n_2	21

The standard deviations of the two samples are unequal. The test statistic is given by:

$$\tau = \frac{\mu_1 - \mu_2}{\sqrt{(\frac{s_1^2}{n_1}) + (\frac{s_2^2}{n_2})}}$$

so

$$\tau = \frac{92 - 91.45}{\sqrt{(\frac{0.4^2}{23}) + (\frac{0.25^2}{21})}} = 5.5185 \qquad \text{to 4dp.}$$

Step 5: Forming a conclusion

You now have:

- Either a Student's t-test or a Welch's t-test.
- Either a one or two-tailed test.
- A null and alternative hypothesis.
- A significance level (α).
- A test statistic (τ).

In order to draw a conclusion for your t test, you have two options:

- Compare the test statistic **directly** to a **critical value.**
- Convert the test statistic to a p-**value.**

> **!** Important
>
> To use **either** approach, you will be asked for the **degrees of freedom**. For all types of t-testing the degrees of freedom are $n - p$.
>
> - n is the sample size.
> - p is the number of samples.

Comparing the test statistic directly to a critical value (t)

Because all t-tests assume that the test statistic follows a t-distribution. You can use a critical value (t) and compare it to your test statistic (τ).

> **i** Summary of critical value method
>
> Step 1: Calculate your test statistic
> Step 2: Calculate critical value
> If your test statistic lies outside of $[-t, t]$ then you **reject** H_0.

Converting the test statistic to a p-value

> **i** Summary of p-value method
>
> Step 1: Calculate your test statistic.
> Step 2: Convert to a p-value.
> Step 3: Compare p-value to the level of significance.
> If your test statistic **less** than your p-value then you **reject** H_0.

Step 6: Summarize your findings in context

Once you have formed your conclusion, you can explain what this means for your scenario or study; putting the t-test in context. You write a concluding statement to say what rejecting, or failing to reject, the null hypothesis means.

For example, in the example where Cantor's Confectionery were testing whether the difference in weight between their Easter eggs to the market average was significant, H_0 suggested no significant difference and H_1 suggested significant difference. If you were to reject H_0 that means there is enough evidence to suggest a significant difference and so the statement would read:

"There is sufficient evidence to reject H_0 meaning there is significant evidence to suggest that the average weight of an Easter egg produced by Cantor's Confectionery differs significantly from the market average."

> **Suggests does not mean total certainty**
>
> In your concluding statement, do not state the results as 100% certain. The results of the test suggests a likelihood so make sure to use the phrases:
>
> - "There is enough evidence to suggest..."
> - "There is not enough evidence to suggest..."

Worked examples

Example 13

Cantor's Confectionery want to perform a t-test, at 5% significance level, to see whether their famous chocolate swirls weigh, on average, the same as the market average of 5.12 grams. The take a sample of 25 swirls with average weight of 4.22 grams and sample standard deviation of 1.23 grams.

Step 1: What type of t-test?

- This is a one-sample test meaning it is a Student's t-test.
- They are testing whether the weight is equal to the market average meaning it is a two-tailed test.

Step 2: Define your null and alternative hypothesis

$$H_0 : \bar{x} = \mu_0 \quad H_1 : \bar{x} \neq \mu_0$$

Step 3: Decide on your significance level (α)

This is a 5% significance level test, so $\alpha = 0.05$.

Step 4: Calculate your test statistic (τ)

For a one-sample Student's t-test the test statistic is

$$\tau = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

so

$$\tau = \frac{4.22 - 5.12}{1.23/\sqrt{25}} = -3.6585 \qquad \text{to 4dp.}$$

Step 5: Forming a conclusion (either approach can be used)

As $n = 25$ and this is a one-sample test, the degrees of freedom are $25 - 1 = 24$.

Critical value approach:

- Critical value calculator gives $t = 1.711$
- Test statistic gives $\tau = -3.6585$
- -3.6585 lies outside of range $[-1.7110, 1.7110]$

so H_0 is rejected.

Example 14

Lovelace's Lollies decide to test whether their chocolate swirls weigh more, on average, than the chocolate swirls made by Cantor's Confectionery. They decide to also use a 5% significance level. They use a computer to produce the following summary statistics:

	Cantor's Confectionery	Lovelace's Lollies
Sample mean	$\mu_1 = 4.22$	$\mu_2 = 3.76$
Sample standard deviation	$s_1 = 1.23$	$s_2 = 1.39$
Sample size	25	25

Step 1: What type of t-test?

- This is a two-sampled test with two different sample standard deviations meaning unequal variance, this is a Welch's t-test.
- They are testing whether the weight is more than Cantor's Confectionery, this is a one-tailed test.

Step 2: Define your null and alternative hypothesis

$$H_0 : \mu_1 = \mu_2 \quad H_1 : \mu_1 < \mu_2$$

Step 3: Decide on your significance level (α)

$$\alpha = 0.05$$

Step 4: Calculate your test statistic (τ)

For a one-sample Student's t-test the test statistic is

$$\tau = \frac{\mu_1 - \mu_2}{\sqrt{\left(\frac{s_1^2}{n_1}\right) + \left(\frac{s_2^2}{n_2}\right)}}$$

$$\tau = \frac{4.22 - 3.76}{\sqrt{\left(\frac{1.23^2}{25}\right) + \left(\frac{1.39^2}{25}\right)}} = 1.2392 \qquad \text{to 4dp.}$$

Quick check problems

1. Which of the following is an appropriate null hypothesis for a one-sample t-test?

(a) $H_0 : \mu_1 > \mu_2$

(b) $H_0 : \mu_1 = \mu_2$

(c) $H_0 : \mu = 81$

2. What is the *rule of thumb* sample size for a t-test?

3. What test would you use to compare means when you have two samples and your data is independent, with different variances?

4. You are performing a one-sample t-test with $H_0 : \mu = 10$. Do you perform a Student's t-test or a Welch's t-test?

Further reading

For more questions on the subject, please go to Questions: Introduction to t-testing.

[For more information on other types of testing, see Guide: More on statistical testing]

Version history

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